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The influence of accounting computer information processing technology on enterprise internal control under panel data simultaneous equation

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Abstract

In order to explore the effect of accounting computer information processing technology on enterprise internal control, this paper puts forward the method of simultaneous equation of panel data. This paper constructs an accounting computer index to measure the accounting computer level of manufacturing enterprises in Shanghai stock market, and studies the relationship between enterprise accounting computer and internal control by using panel data simultaneous equations model. The empirical results show that: first, although the average accounting computer level of manufacturing industry increases year by year, the overall level is low, and there are great differences among sub industries, even within the same industry; Second, the level of accounting computer has a negative impact on the internal control of enterprises, and the internal control of enterprises has a positive impact on the level of accounting computer; Third, enterprise scale and operating years have a significant positive impact on the level of accounting computer. The experimental results show that: from the estimation results of accounting computer equation, firstly, ROA has a significant positive impact on the level of accounting computer; Secondly, the scale and operating age of enterprises positively affect the level of accounting computer, while the nature of ownership and listing age are not related to the level of accounting computer; Finally, document disclosure has a high level of accounting computer, which is consistent with the conclusion of the above descriptive statistics of EMI, which may be related to the mandatory provisions of national laws and regulations on the industry of document disclosure.

Keywords: Panel data simultaneous equation; Computer information processing technology; Enterprise internal control
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1. Introduction

As the final product of daily accounting, accounting information directly reflects the performance ability of enterprises and affects the national macro-control. According to the relevant economic models, accounting information accounts for as much as 70% of the total amount of enterprise information. Investors can indirectly analyze the solvency, operating capacity, market performance and other comprehensive indicators of enterprises through their various data. Therefore, enterprises with true and accurate accounting information will undoubtedly attract the attention of investors and shareholders. In the traditional accounting environment, although enterprises have also established relevant management information systems, due to the single structure of their own internal control and many factors of the external environment, they fail to give full play to the economic benefits of accounting information. With the popularization of the era of big data, the quality of accounting information has been sublimated in all aspects[1]. The author mainly analyzes the current advantages of accounting information from the perspectives of information output, information acquisition and information operation cost. Under the traditional accounting environment, the output of enterprise accounting information has obvious delay. In some repetitive and complicated work, the audit, signature, reconciliation, settlement and other work links are mainly completed by accountants in the way of "manual recording". The time cost of coordination between them is relatively large and the business completion efficiency is relatively low. The main problem is that the staff did not realize the maximum advantage of "1 + 1 > 2", and finally failed to make full use of the information resources owned by the enterprise[2]. Under the background of big data information age, enterprises mainly implement computerized accounting and "ERP" mode to carry out unified work management of resources, logistics and information owned by enterprises, and are no longer constrained by the limitation of "working distance" under the traditional accounting mode. It allows every accountant to share the data to be processed through the accounting cloud system platform, and further integrate the high-quality accounting information of the final output[3].

After entering the 21st century, information technology has brought new impetus to the development of enterprises. Many enterprises have gained new development with the support of information technology. Therefore, enterprises attach more importance to information technology. Information technology has been adopted by more and more enterprises, and the core competition of enterprises has changed from product to information technology. The internal environment of enterprises has also changed with the arrival of information technology. Modern enterprise management has changed in time and space, and the organizational existence form, business model and humanistic environment of enterprises have been innovated. The information technology capability formed by the combination of information technology with strategy, organization, personnel and process is not only difficult to be imitated, but also becomes the focus of continuous competition in the future. At the same time, information technology capability will rely on the above parts to penetrate into the whole process of enterprise management, and constantly affect the internal control of enterprises. Before enterprises use information technology, enterprise internal control is people-oriented, in accordance with the internal control standards and accounting standards for enterprises, the control of employees, business, to ensure the integrity of information. However, the level of internal control under person-to-person control is difficult to guarantee, which will affect the quality of disclosed information. After the combination of information technology and internal control, the malpractice of "people control people" has been solved, and the integration of internal control rules and information system will become a new problem. The enterprise's control environment, risk assessment, control activities, information and communication, and monitoring have been changed by information technology capabilities. Information technology provides the possibility for improving the level of enterprise internal control, and the level of information technology capability becomes the key to effectively implement internal control. At the same time, the risks brought by information technology will be manifested through information technology capability. Obviously, information technology ability will have a profound impact on the enterprise internal control, then the level of information technology ability on the enterprise internal control level will have what impact and how information technology ability affects the enterprise internal control? This paper will study the above problems through normative analysis, empirical analysis and case analysis.

Deng Jdivided information technology into four levels, including relevant training modes and management means in information management; Application of modes and means; Interaction between personnel and computers; Corresponding social, economic and cultural factors. To sum up, the role of information technology needs a certain carrier, and the process of integration with the carrier is informatization. Scholars at home and abroad have different opinions on the definition of informatization[4]. Ota T A

believes that informatization is economic informatization based on enterprises and market, government informatization based on government and social informatization based on people's daily life[5]. Trushkevych O believes that different scholars have different definitions of informatization, but they are consistent in the basic purpose, scope, implementation means and process[6]. Based on this, he puts forward that the purpose of informatization is to improve the core advantages of enterprises, so as to enhance the market share.

Based on the current research, this paper proposes a simultaneous equation method for panel data. This paper constructs an accounting computer index to measure the accounting computer level of manufacturing enterprises in Shanghai stock market, and studies the relationship between enterprise accounting computer and internal control by using panel data simultaneous equations model. The empirical results show that: first, although the average accounting computer level of manufacturing industry increases year by year, the overall level is low, and there are great differences among sub industries, even within the same industry; Second, the level of accounting computer has a negative impact on the internal control of enterprises, and the internal control of enterprises has a positive impact on the level of accounting computer; Third, enterprise scale and operating years have a significant positive impact on the level of accounting computer.

2. Construction of accounting computer index

This paper systematically constructs the accounting computer level rating index system from the following three aspects. In terms of internal management of enterprises, the accounting computer of enterprises should be carried out from inside to outside, from top to bottom, from management to employees, from management system to final product, which should reflect the concept of accounting computer. First, the management should make a commitment to environmental protection in the process of operation and development, and incorporate environmental protection into the future development plan of the enterprise, which is the premise for the enterprise to fulfill its environmental responsibility[7]. Second, accounting computer also needs the participation of employees. The enterprise shall conduct regular environmental protection training for employees to improve the organization's environmental awareness. Third, enterprises should set up special environmental prevention and treatment departments. The management mode of pollution prevention reflects a higher level of accounting computer than the traditional terminal treatment mode. Fourth, enterprises should increase the production and technology research and development of environment-friendly products, which will help reduce the negative impact of enterprises on the environment and reflect the specific investment and expenditure of enterprises in accounting computers. In terms of external management, on the one hand, enterprises should timely and accurately disclose social responsibility or environmental reports, on the other hand, they should communicate and cooperate with domestic and foreign green environmental protection enterprises. In terms of external recognition, enterprise accounting computer achievements should be recognized by the outside world. Accounting computer standards, such as ISO14000 series standard certification, on the one hand, as a tool of accounting computer, on the other hand, it is a sign of the level of accounting computer. The certification of accounting computer standards and environmental protection awards show that the accounting computer behavior of enterprises has been objectively evaluated by the public.

Therefore, the construction of accounting computer index (hereinafter referred to as EMI) in this paper includes three aspects, a total of 10 indicators, and the specific connotation is shown in Table 1. Since China has not implemented the public disclosure system of environmental reports, nor has it formed an open and systematic accounting computer evaluation system, the available information can only be obtained from the enterprise annual report or social responsibility report. For example, the relevant indicators of "internal management" and "external recognition" generally appear in the "report of the board of directors"[8]. This paper searches the annual report or social responsibility report of the enterprise according to the above 10 indicators. If it is found that the enterprise activities meet one of the indicators, the score of the accounting computer index will be increased by 1. Therefore, the accounting computer index (EMI) obtained in this paper is an ordered variable with a value between 0 and 10. A higher EMI reflects a higher level of accounting computer.

Table 1 Management index used in this paper

Content of management	Management capability metrics

Internal management	management	Are there any explanations for environmental protection in future development goals?
		Whether specific strategies and measures have been approved
	staff layer	Are employees regularly trained and educated on the environment?
	management organizational structure	Are employees regularly trained and educated on the environment?
	Green technology research and development	Whether there is a product process-friendly product design and development
External management	Information disclosure	Can a social responsibility report be issued in a timely manner?
	Cooperation	Ability to sponsor, support, and collaborate with environmental agencies
		Is it possible to communicate and cooperate with domestic and foreign green environmental protection companies?
External recognition	Certification	Whether it exceeds the IS014000 management system certificate
	Reward situation	Whether to receive protection bonuses during use

Defects exposed by enterprise internal control under traditional accounting under traditional work, many entrepreneurs believe that internal control is static, independent and irrelevant to the outside world, resulting in many business problems. For example, incomplete disclosure of financial statement information, misappropriation of public funds by accountants, frequent implementation of personal opportunism by managers and so on, eventually make the enterprise face the outcome of bankruptcy and delisting. The author mainly analyzes the problems of enterprise internal control under traditional accounting from four main aspects: accounting personnel, internal supervision, information security and performance management[9].

(1) The professional ability and professional ethics of accountants need to be improved. Under the traditional work, accountants only pay attention to the compliance of account books and the completeness of documents in order to complete the expected business volume, and do not learn to use existing resources to improve the company's performance in the market. For some overseas M & A enterprises, accountants cannot skillfully operate the financial information system, which makes the business slow and the performance not up to standard, resulting in a large number of customer chain breaks, and finally bring serious economic losses to the enterprise. In addition, many investment businesses have greater flexibility and randomness. In order to maximize their personal interests, enterprise managers use the raised shareholder capital to tend to the change of strategic objectives, and fail to weigh the opportunity cost risk and tax related risk of the project, which increases the uncertainty of the long-term development of the enterprise, which is seriously contrary to the professional ethics of accountants[10].

(2) The internal discipline inspection and supervision mechanism lags behind. Although enterprises have established a relatively complete theoretical system of internal control, the supervision mechanism is not complete, resulting in many "missed fish" cases. In order to whitewash the financial statements and falsely increase profits, some operators have reached a "network of acquaintances" relationship with auditors, and have not combined their personal interests with the interests of the company at all. After the field investigation of several listed companies, the author summarizes the following problems:
①There is functional compatibility between business decision-makers and business executives;
②Failure to give timely feedback on major matters in the business year of the enterprise;
③There is no clear and unified standard for the delivery pricing method of inventory;
④The time of property inventory is not clear enough;
⑤Most of the auditors are financial personnel, with insufficient reserves of relevant professional knowledge and lack of work experience, resulting in the failure to refine their daily work;
⑥Over reliance on business personnel and enterprise resources in the hands of individuals are not conducive to the overall management of business;
⑦There are many descriptive provisions in the internal control system, and there are few clear flow charts and supporting tables.

(3) The defense ability of enterprise information security is not optimistic. Information is a business opportunity. It may bring potential opportunities or unknown manipulation risks to enterprises. With the disclosure of core information and the tampering of business strategies, more and more criminal groups began to plagiarize the labor achievements of enterprises and drill loopholes in internal control, resulting in a large number of information security accidents. According to the 18th global information security survey report, among 1755 invited enterprises, 20% of respondents said that there had been many information security problems in various departments of enterprises in the recent fiscal year; 88% of respondents said that the current information security capability of enterprises did not meet the expected development needs; 36% of respondents said they did not understand the enterprise's smart threat system; 47% of respondents said that enterprises did not install a firewall system platform to deal with information security problems; 17% of the enterprises surveyed said that they would face the problem of information security again in the next three years. It can be seen that in the traditional accounting environment, enterprise employees have a vague understanding of information security and fail to combine information security with property security, resulting in a large number of unexpected losses of resources.

(4) Enterprise performance evaluation standards are relatively vague. With the in-depth development of China's socialist market economy theoretical system, enterprises have more initiative and business activities have more regularity. However, so far, domestic enterprises still lack a set of scientific and complete enterprise performance evaluation system, and there are certain limitations and lags in the design of performance indicators. Comparing the operating level of enterprises in different industries and sizes with the same indicators will bring wrong information guidance and judgment to enterprises, which will seriously hinder the construction of internal control[11]. The author summarizes the common performance management problems as follows:
① In the initial stage, some enterprises only measured the company's performance by market interests, shareholder wealth and other indicators, without considering the long-term indicators such as the enterprise's sustainable development ability and risk aversion flexibility. ② Decision makers covet immediate interests and give priority to projects with short payback period, but give up projects with higher return on investment in the future, resulting in many short-term behaviors. ③ Some immature enterprises believe that the higher the profitability, the better. A large number of loans are invested in economic business, completely ignoring the solvency of the enterprise. Finally, they are insolvent and face bankruptcy.

3 Sample selection and model construction

Sample selection in order to eliminate the influence of industry factors, according to the industry classification of the China Securities Regulatory Commission, this paper selects the manufacturing enterprises listed on the Shanghai Stock Exchange as the sample, excludes the companies that

simultaneously issue B shares, and excludes st enterprises and newly listed or newly restructured enterprises in these six years, and obtains the panel data group of 305 enterprises from 2007 to 2012. Enterprise internal control. According to the selected indicators of enterprise internal control measurement, combined with the weak effectiveness of China's capital market, Tobin Q can not accurately reflect the current financial performance of the enterprise. Therefore, this paper selects two indicators: return on net assets (ROE) and return on total assets (ROA) to measure enterprise internal control. Enterprise characteristic variables. According to the characteristics of enterprises, scale, joint-stock nature and operation time have a significant impact on the relationship between corporate social responsibility and internal control. First of all, large-scale enterprises will be strictly supervised by public opinion and have more external pressure to improve their accounting computer level[12]. At the same time, the advantage of scale makes it more likely to invest in the practice of accounting computer. Secondly, the nature of corporate joint-stock system will affect investors' cognitive attitude towards responsibility. Due to China's current economic system with public ownership as the main body, there is a close relationship between state-owned enterprises and the government, which is more vulnerable to preferential treatment by the government, or lead to insufficient investment in environmental protection and governance, and then ignore the internal accounting computer. Finally, enterprises that have operated for a long time have more operation and management experience and more management talents to practice environmental protection than young enterprises. The listing of enterprises can absorb idle funds from the society for financing, and there will be more abundant resources for technological innovation, updating and transforming the original equipment and producing environment-friendly products. Other variables. Enterprise internal control may be affected by many factors, and the influence of relevant variables must be controlled.

The internal control equation and accounting computer equation are expressed as follows:
Enterprise internal control equation:

$$\begin{aligned} ROA_{qp} &= \alpha_{1qp} + \alpha_{2qp} \times EMI_{qp} + \alpha_{3qp} \times DA_{qp} \\ &+ \alpha_{4qp} \times CE_{qp} + \alpha_{5qp} \times ADV_{qp} + \alpha_{6qp} \times SIZE_{qp} \end{aligned} \tag{1}$$

$$\begin{aligned} ROE_{qp} &= \alpha_{1qp} + \alpha_{2qp} \times EMI_{qp} + \alpha_{3qp} \times DA_{qp} \\ &+ \alpha_{4qp} \times CE_{qp} + \alpha_{5qp} \times ADV_{qp} + \alpha_{6qp} \times SIZE_{qp} \end{aligned} \tag{2}$$

Enterprise accounting computer equation:

$$\begin{aligned} EMI_{qp} &= \alpha_{7qp} + \alpha_{8qp} \times ROA_{qp} + \alpha_{9qp} \times SIZE_{qp} \\ &+ \alpha_{10qp} \times OWN_{qp} + \alpha_{11qp} \times AGE1_{qp} + \alpha_{12qp} \\ &\times AGE2_{qp} + \alpha_{13qp} \times INDU_{qp} \end{aligned} \tag{3}$$

$$\begin{aligned} EMI_{qp} &= \alpha_{7qp} + \alpha_{8qp} \times ROE_{qp} + \alpha_{9qp} \times SIZE_{qp} \\ &+ \alpha_{10qp} \times OWN_{qp} + \alpha_{11qp} \times AGE1_{qp} + \alpha_{12qp} \\ &\times AGE2_{qp} + \alpha_{13qp} \times INDU_{qp} \end{aligned} \tag{4}$$

Where subscripts q and p respectively represent the company and time in the panel data. $a_n(n = 1,2,3.....)$ represent the coefficients of each variable in the model. Internal control equation (1)

and accounting computer equation (3) are combined into model 1, and internal control equation (2) and accounting computer equation (4) are combined into model 2.

4 Experimental results and analysis

4.1 Enterprise accounting computer level

After testing the endogeneity and simultaneity of the equation, this paper uses three estimation methods to estimate the simultaneous equation. Firstly, the simultaneous equation is estimated by the least square method; Then the simultaneous equations are estimated by the fixed effect two-stage least square method; Finally, the simultaneous equations are systematically estimated by three-stage least square method. The biggest difference between the three-stage least squares method and the two-stage least squares method is that the correlation between the residual terms of different equations is considered, but the two-stage least squares method is generally used for estimation, because the correlation between the residual terms of equations is usually weak. It can be seen from the results that the two-stage least squares method is better than the three-stage least squares method, so the two-stage least squares method is used to estimate the impact of state-owned enterprises. At the same time, because this paper mainly studies the interaction between R & D intensity RDI and salary gap PG on performance, it focuses on the regression of the first equation (the second stage), and the second equation is mainly used to control the endogeneity of salary gap[13].

Firstly, the average accounting computer level of manufacturing industry has increased year by year, but its overall level is still low; Secondly, the average accounting computer level of each sub industry is quite different, and the average accounting computer index of the document disclosure industry is higher than the average level of the whole industry; Thirdly, the range data shows that even in the same industry, the accounting computer indexes of enterprises are far from each other, indicating that enterprises have not reached a consensus on the importance of disclosing relevant information[14].

4.2 Simultaneous equation estimation results of panel data

In this paper, the two-stage least square method (2SLS) is selected for estimation. The estimation results are shown in Table 2 and 3.

1. The influence of enterprise accounting computer on internal control. From the estimation results of internal control equations, firstly, EMI has a significant negative impact on ROA and roe; Secondly, consistent with the conclusions of most literatures, asset liability ratio and operating efficiency negatively affect enterprise internal control, while intangible asset ratio and scale positively affect internal control.

2. The influence of enterprise internal control on enterprise accounting computer. From the estimation results of accounting computer equation, firstly, ROA has a significant positive impact on the level of accounting computer; Secondly, the scale and operating age of enterprises positively affect the level of accounting computer, while the nature of ownership and listing age are not related to the level of accounting computer; Finally, enterprises in the document disclosure industry have a high level of accounting computer, which is consistent with the conclusion of the above descriptive statistics of EMI, which may be related to the mandatory provisions of national laws and regulations on the document disclosure industry[15].

Table 2 Simultaneous equation estimation results of panel data (Model 1)

Model 1			
Technical equation of accounting computer processing		Internal control equation	
	ROA		EMI
	Coefficient		Coefficient

C	-0.1547	C	-17.5313
EMI	-0.0086	BOA	3.3757
DA	-0.0937	SIZE	0.8674
CE	-0.1874	OWN	0.1098
ADV	0.1236	AGE1	0.0533
SIZE	0.0169	AGE2	0.0174
		INDU	0.3263

Table 3 Simultaneous equation estimation results of panel data (Model 2)

Model 2			
Technical equation of accounting computer processing		Internal control equation	
	ROA		EMI
	Coefficient		Coefficient
C	-0.6469	C	-17.2469
EMI	-0.0733	BOA	1.6709
DA	-0.0518	SIZE	0.5818
CE	-0.3367	OWN	0.0689
ADV	0.2865	AGE1	0.0547
SIZE	0.0371	AGE2	0.036
		INDU	0.274

5. Conclusion

This paper systematically constructs the computer processing technology index and quantitatively measures the computer processing technology level of listed manufacturing enterprises in Shanghai Stock Exchange. It is found that although the average computer processing technology level of manufacturing industry increases year by year, the overall level is low, and there are great differences among sub industries, even within the same industry. Using the simultaneous equation model of panel data, this paper makes an empirical study on the relationship between enterprise computer processing technology and internal control under the control of enterprise industry, scale and other factors. The conclusion shows that the computer processing technology level of enterprises negatively affects the internal control, while the internal control, scale and operation years positively affect the computer processing technology level. The nature of ownership and listing years have no correlation with the computer processing technology level. The advent of the era of big data provides a guarantee for the accounting information of enterprises. Efficient accounting information will not only attract the attention of more investors, but also inject new vitality into the internal control of enterprises. We are well aware that the construction of internal control is a dynamic process that keeps pace with the times. With the continuous improvement of the enterprise's own factors, the system of internal control is also changing imperceptibly. Internal control is not only for the management or the board of directors, but also for all employees of the enterprise. Therefore, we advocate the thinking mode of combining personal interests with the interests of the company, which helps each employee take the initiative to undertake more work responsibilities, which will not only bring stronger discipline and safety to the enterprise, but also eliminate many cases of corruption. The author firmly believes that under the guidance of the era of big data, China's internal control system and enterprise

development will go to a higher level.

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